

Experiment 06 — Publication Audit Addendum

Post-hoc methodological review | Experimental computational model | Not peer-reviewed

What the original simulation demonstrates

Across 24 independent evolution seeds, selected recurrent controllers achieved mean exact expected payoff 0.1705 versus 0.0833 under random-fitness selection (paired mean difference 0.0872; 95% bootstrap interval 0.0611–0.1130). This demonstrates the usefulness of a mutation-enabled action-history link in a deliberately constructed environment. It does not measure consciousness or a genuine self-model.

Important control flaw

The original no-inheritance condition also prevents recurrent-link mutation. It cannot isolate the effect of inheritance. Its zero-score outcome must not be interpreted as an independent proof that inheritance causes self-modeling.

Post-hoc follow-up baseline: fresh genomes with possible links

A new, separately labelled sensitivity check regenerated 192 unrelated agents in each generation and allowed each fresh genotype to have recurrent links. Selection of the best final-generation controller used the same analytic survival criterion. This deliberately simplified control is not a rerun of the original evolutionary process.

Condition	Mean payoff	95% bootstrap interval
Original evolved	0.17051	0.14865–0.18812
NEW de novo no-inheritance	0.02030	0.01766–0.02323
Paired difference	0.15022	0.12943–0.16933

Additional weaknesses

Training optimizes exact payoff using the known environment distribution; the motor command is externally supplied rather than chosen; champions are chosen post hoc by highest true task payoff even for the random-fitness control; the task and possible recurrent architecture are engineered; there is no independent replication or public preregistration.

Release recommendation

Share as an exploratory model only, with all code and the methodological addendum. For academic submission, preregister a new ecological simulation with stochastic selection, agents choosing their actions, clean mutation and inheritance controls, multiple architecture families, and independent external replication.

Research context

Crapse & Sommer (2008), DOI: 10.1038/nrn2457; Bongard et al. (2006), DOI: 10.1126/science.1133687; Tanaka & Imamizu (2025), DOI: 10.1038/s44271-025-00240-7. The general phenomenon predates this simulation.